Liam Meagher

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PROFESSIONAL SUMMARY

Mechanical engineer with a strong foundation in mechanical design, fabrication, and systems integration, backed by hands-on experience in CAD modeling, FEA, GD&T, and CNC manufacturing. Combines hardware expertise with autonomous systems software development, enabling end-to-end ownership of complex engineering projects from concept through physical validation.

EDUCATION

University of Southern California
M.S. in Mechanical Engineering — *Expected May 2028*

University of California, Los Angeles
B.S. in Computer Science & Engineering — *September 2021 - June 2025*

WORK EXPERIENCE

Computer Engineering Intern — **August 2026 – Present**
Copy Solutions Inc. — *Alhambra, CA*

- Developed FPGA solutions for industrial applications and manufacturing workflows using Zebra and Xerox technologies
- Optimized DPM processes, data matrices, and barcodes for improved contrast, readability, and Zebra scanner reliability
- Integrated RFID, BLE, and UHF sensing technologies for inventory tracking and cradle-to-grave product traceability

Operations Management Intern — **March 2026 – August 2026**
Scouting America — *Avalon, CA*

- Improved operational systems supporting outdoor programs serving over 4,000 participants with a $390K+ annual budget
- Engineered new inventory management systems for 220+ staff, improving asset visibility and reducing supply errors by 15%
- Built financial reports and performance dashboards to evaluate 16 concurrent projects and optimize resource utilization
- Identified process bottlenecks and implemented workflow improvements that increased operational efficiency
- Led training initiatives for all employees, developing standardized documentation and procedures to improve consistency

Engineering Instructor & Product Development Mentor — **August 2025 – May 2026**
Heart of Los Angeles — *Los Angeles, CA*

- Coached student engineering teams through end-to-end product development processes, creating functional prototypes
- Led iterative design cycles to define user requirements, prioritize features, and evaluate engineering tradeoffs
- Oversaw technical projects from concept generation through prototype testing, ensuring projects met performance objectives
- Mentored students in CAD, rapid prototyping, design reviews, and technical presentations emphasizing improvement

TECHNICAL PROJECTS

Chassis Lead Engineer — **January 2023 – June 2025**
Flagship Combat Robotics — *Los Angeles, CA*

- Led 30+ team members through the research, design, and fabrication of 4 combat robots with various spinner weapon systems
- Applied GD&T and tolerance stack-up analysis to achieve precise fits, alignment, and 20+ DFM-ready component designs
- Used SolidWorks CAD and FEA to validate the design integrity of chassis subassemblies under high-shock loading (5 kN)
- Employed Ansys Discovery topological optimization to reduce mass by 42% (10.3 lb) while maintaining stiffness targets
- Created over 50 fabrication drawings and release packages with datums, tolerances, and inspection-critical dimensions
- Accelerated chassis and drivetrain prototyping cycles by 30% through rapid iteration with 3D printing and laser cutting

Autonomous Systems Software Developer — **September 2023 – June 2024**
Bruin Underwater Robotics — *Los Angeles, CA*

- Developed autonomy control software for an underwater robot, implementing real-time feedback control loops for depth
- Improved closed-loop stability by reducing oscillations by 40% through PID tuning and dynamic response testing
- Generated 2000+ simulated sensor frames and produced 5+ hours of synthetic mission data using Unity's simulation tools
- Conducted simulation-driven reliability testing across 8 mission scenarios, increasing autonomous control stability by 45%
- Integrated control software with thrusters, depth sensors, and embedded hardware to enhance performance and reliability during autonomous underwater pathing missions using technologies such as ROS and C++

SKILLS

Software: SolidWorks (CAD/FEA), Onshape (CAD/PDM), Ansys Discovery, MATLAB, Linux, Excel, Unity, Unreal Engine
Manufacturing: CNC (Mill/Lathe), 3D Printing (FDM), Lean Six Sigma, GD&T, Design for Manufacturability (DFM)
Languages: C, C++, C#, Haskell, Java, JavaScript, Python, R, RISC-V Assembly, SQL, Verilog
Libraries & Tools: NumPy, Pandas, Matplotlib, OpenCV, Scikit-Learn, PyTorch, React, Node.js, Docker, Git, Arduino, ROS 2